STAT A411/611 Lecture Notes

Instructor: Rieken Venema **Office:** SSB 158E
Office hours: MWF 12:00–12:45 **Class:** 2:30–3:45

Lecture Plan and Review

Today:	Chapter 1, Sections 1.1, 1.2, and 1.3.
Monday:	No class.
Wednesday:	Continue Section 1.3.

Review of Section 6.4

The main topics were z-scores, percentiles, quartiles, outliers, and boxplots. A boxplot is built from ordered data using Q_1, Q_2, Q_3, the IQR, the outlier check, and the whisker endpoints P_1 and P_2. Its five-number summary is

$$\{P_1, Q_1, Q_2, Q_3, P_2\},$$

with outliers plotted separately.

1 Chapter 1 — Probability

1.1 Section 1.1 — Classical Formula for Probability Calculations (C.F.)

Sample Space

An *experiment* is performed and its possible outcomes are collected in a set.

Definition. The *sample space* is the set of all possible outcomes of an experiment. Its notation is S.

Example: two dice. Roll a fair six-sided red die and a fair six-sided blue die and observe the sum of the two upward-facing sides. Then

$$S = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}.$$

Events

Definition. Any subset of S is called an *event*. If the observed outcome falls in the set A, then event A occurs. An event can therefore be described verbally or written as a set.

For the two-dice experiment,

$$A = \{\text{roll an even sum}\} = \{2, 4, 6, 8, 10, 12\}.$$

Probability Notation

$P(A)$ denotes the probability that event A occurs, or how likely it is that the outcome of the experiment falls in set A. Its value always satisfies

$$\boxed{0 \le P(A) \le 1}.$$

If $P(A) = 0$, A is an impossible event; if $P(A) = 1$, A is a certain event. For example, with two ordinary dice,

$$P(\text{sum} < 2) = 0, \qquad P(\text{sum} \le 12) = 1.$$

Classical Formula for Probability Calculations

Suppose all outcomes in S are equally likely. Then for any event A,

$$\boxed{P(A) = \frac{\#\{\text{elements in } A\}}{\#\{\text{elements in } S\}}} = \frac{\#\{\text{ways event } A \text{ can occur}\}}{\#\{\text{possible outcomes}\}}.$$

Example 1: Flip a Fair Coin Twice

What is the probability that at most one head comes up? The sample space is

$$S = \{HH, HT, TH, TT\},$$

and all four outcomes are equally likely. Hence

$$\boxed{P(\text{at most one head}) = \frac{3}{4}}.$$

Example 2: Sum of a Red and Blue Die

A first attempt is

$$P(\text{sum} = 8) = ?, \qquad S = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}.$$

If the Classical Formula were applied directly to these eleven values, it would give

$$P(\text{sum} = 8) = \frac{1}{11}.$$

This attempt fails because the eleven possible sums are not equally likely. For example, a sum of 2 has one ordered realization, while a sum of 7 has six.

Second, better attempt. Use the 36 equally likely ordered pairs (R, B).

$R\backslash B$	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	**8**
3	4	5	6	7	**8**	9
4	5	6	7	**8**	9	10
5	6	7	**8**	9	10	11
6	7	**8**	9	10	11	12

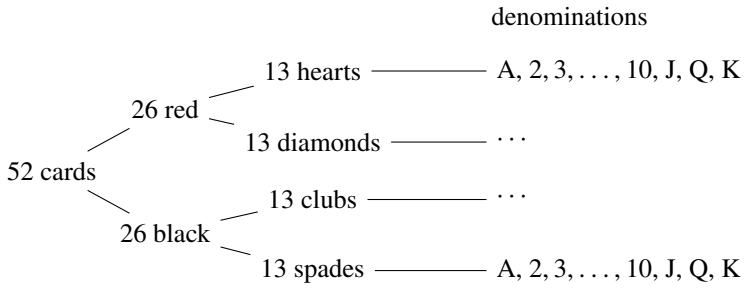
R	B
2	6
3	5
4	4
5	3
6	2

There are five equally likely pairs giving a sum of 8, so

$$\boxed{P(\text{sum} = 8) = \frac{5}{36}}.$$

A Deck of 52 Playing Cards

A standard deck contains 52 cards:



Each suit contains the denominations ace, 2, 3, . . . , 10, jack, queen, and king. Therefore,

$$P(\text{red}) = \frac{26}{52} = \frac{1}{2}, \qquad P(\text{spade}) = \frac{13}{52} = \frac{1}{4}, \qquad P(\text{a 7}) = \frac{4}{52} = \frac{1}{13}.$$

Poker Setup

A dealer shuffles a deck of 52 cards and gives a player 5 randomly selected cards without replacement. For the event "exactly two spades," the classical-formula setup is

$$P(\text{exactly two spades}) = \frac{\#\{\text{5-card selections with 2 spades and 3 non-spades}\}}{\#\{\text{5-card selections}\}}.$$

The counting tools needed to evaluate this ratio are developed in Section 1.3.

Homework: 2 and 4 (pp. 17–18).

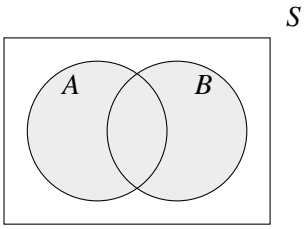
1.2 Section 1.2 — Some Concepts from Set Theory

Union and Intersection

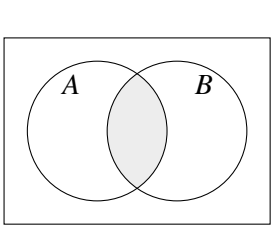
For two events A and B:

- The *union* $A \cup B$ occurs if either A occurs or B occurs (or both). Equivalently, at least one of the two events occurs.
- The *intersection* $A \cap B$ occurs if both A and B occur.

$$A \cup B = A \text{ OR } B, \qquad A \cap B = A \text{ AND } B.$$



$A \cup B$



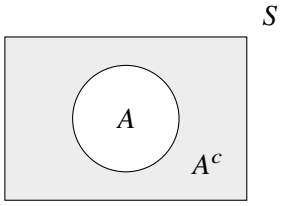
$A \cap B$

For several events,

$$\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \cdots \cup A_n, \qquad \bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \cdots \cap A_n.$$

Complement

The *complement* of A, denoted A^c (written A' in the textbook), occurs when A does not occur.

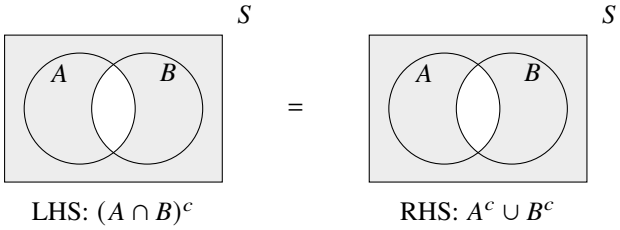


The basic properties are

$$A \cap A^c = \varnothing, \qquad A \cup A^c = S.$$

De Morgan's laws are

$$\boxed{(A \cap B)^c = A^c \cup B^c}, \qquad \boxed{(A \cup B)^c = A^c \cap B^c}.$$



LHS: $(A \cap B)^c$ RHS: $A^c \cup B^c$

Mutually Exclusive Events

Two events A and B are *mutually exclusive* (or *disjoint*) if they cannot occur at the same time:

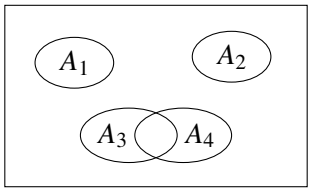
$$\boxed{A \cap B = \varnothing}.$$

Equivalently, at most one of them can occur.

More generally, $A_1, A_2, \ldots, A_n$ are mutually exclusive if

$$\boxed{A_i \cap A_j = \varnothing \quad \text{for all } i \neq j},$$

where $i, j \in \{1, 2, \ldots, n\}$ index the events.



In this configuration, A_1, A_2, A_3 are mutually exclusive and A_1, A_2, A_4 are mutually exclusive, but all four events are not mutually exclusive because A_3 and A_4 overlap.

Text reference: p. 9, Definition 1.2.3. **Homework:** 6 and 8.

1.3 Section 1.3 — Counting Rules

Combinatorial Analysis

I. Basic Rule of Counting (BRC)

Suppose we perform k successive experiments. If experiment i has n_i possible outcomes, for $i = 1, 2, \ldots, k$, then the entire sequence has

$$\boxed{n_1 n_2 \cdots n_k}$$

possible different outcomes.

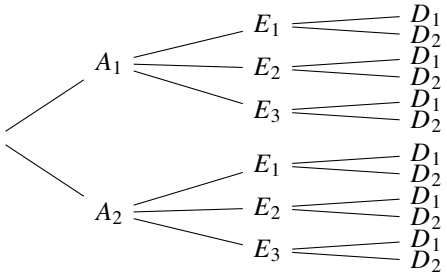
This is the counting rule used to determine the numerator and denominator in classical probability calculations when direct listing is impractical.

Example 1: Dinner Menu

A restaurant dinner consists of an appetizer, an entree, and a dessert. If there are 2 appetizers, 3 entrees, and 2 desserts, then

$$\boxed{2 \cdot 3 \cdot 2 = 12}$$

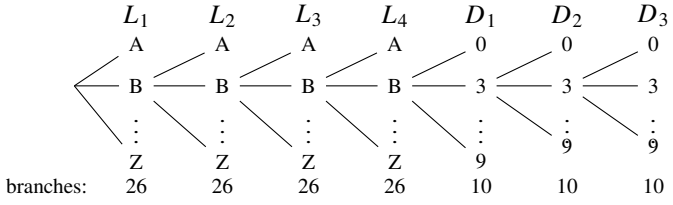
different dinners are possible. The successive choices can be represented by a tree:



Example 2: License Plates

A license plate consists of four letters (A–Z) followed by three digits (0–9). Write the letter positions as L_1, L_2, L_3, L_4 and the digit positions as D_1, D_2, D_3, where L denotes a letter and D denotes a digit.

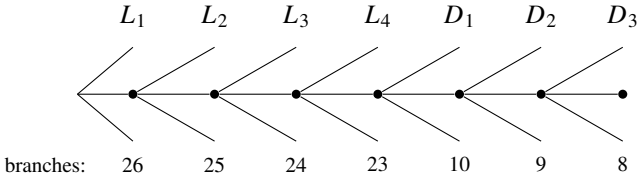
With repetition. A schematic choice tree is



Thus, by the Basic Rule of Counting,

$$\boxed{26 \cdot 26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 = 26^4\, 10^3}.$$

Without repetition. If repetition is not allowed, each successive letter choice leaves one fewer letter available, and each successive digit choice leaves one fewer digit available:



Therefore,

$$\boxed{26 \cdot 25 \cdot 24 \cdot 23 \cdot 10 \cdot 9 \cdot 8}.$$

Likewise, the BRC can be used to count the number of possible passwords, ZIP codes, telephone numbers, and similar sequential constructions.

II. Permutations

Start with n distinct elements. Select k elements randomly and without replacement, where

$$1 \le k \le n,$$

and where the *order of selection matters*. Every different ordered selection is called a *permutation*.

The total number of possible permutations is denoted ${}_nP_k$. To find a formula using the BRC, view the selection as k successive choices:

$$\begin{array}{ccccccc} \text{sel}_1 & & \text{sel}_2 & & \cdots & & \text{sel}_k \\ n & \longrightarrow & n-1 & \longrightarrow & \cdots & \longrightarrow & n-k+1 \end{array}$$

Hence,

$${}_nP_k = n(n-1)(n-2)\cdots(n-k+1).$$

For example,

$${}_nP_2 = n(n-1), \qquad {}_nP_3 = n(n-1)(n-2).$$

Factorials

Zero factorial is defined to be 1. Thus,

$$0! = 1, \qquad 1! = 1, \qquad 2! = 2 \cdot 1 = 2, \qquad 3! = 3 \cdot 2 \cdot 1 = 6,$$

and, for example,

$$4! = 24, \qquad 5! = 120, \qquad 6! = 720, \qquad 10! = 3{,}628{,}800 > 3.5 \text{ million}.$$

To rewrite the permutation formula in factorial form, extend the product by the remaining factors and divide by the same quantity:

$$\begin{aligned} {}_nP_k &= n(n-1)\cdots(n-k+1) \\ &= \frac{n(n-1)\cdots(n-k+1)(n-k)(n-k-1)\cdots 3\cdot 2\cdot 1}{(n-k)(n-k-1)\cdots 3\cdot 2\cdot 1} \\ &= \boxed{\frac{n!}{(n-k)!}}. \end{aligned}$$

Combinations

A related formula for combinations is

$$\boxed{{}_nC_k = \binom{n}{k} = \frac{n!}{k!(n-k)!}}.$$

This topic will be developed later.

Next lecture: continue Section 1.3.